

MODEL VALIDATION REPORT

Independent Model Risk Management (MRM) review — for credit risk committee sign-off

Model: Retail PD Scorecard (Lending Club) | Owner: Credit Risk Analytics | Validator: MRM | Date: 23 Jun 2026

1. Model scope & methodology

Application-scorecard estimating 12-month Probability of Default (PD) on unsecured consumer loans. Method: Weight-of-Evidence (WoE) binning + multivariate logistic regression (statsmodels MLE), scaled to a points scorecard (PDO=20, base=528). PD feeds an Expected-Loss framework $ECL = PD \times LGD \times EAD$; LGD/EAD are LightGBM regressors on the defaulted population. A LightGBM model serves as the non-linear challenger benchmark.

2. Data & sampling

Source: Lending Club accepted loans 2007-2018 (1,345,350 performance loans, default ~20%). Target: Charged Off / Default = 1; Fully Paid = 0; in-flight statuses excluded. Development cohort (issued ≤ 2016): 1,119,711 loans, fit on a 70/30 stratified split. Out-of-time (OOT) cohort (2017-2018): 225,639 loans, reserved for stability testing.

3. Discriminatory power (out-of-sample test set)

Metric	Champion (Scorecard)	Challenger (LightGBM)	Gate	Result
AUC	0.7145	0.7210	-	-
Gini	0.4291	0.4420	-	-
KS %	31.07	32.06	≥ 30	PASS

4. Key validation findings

LEAKAGE CONTROL total_rec_prncp (IV~2.18) and recoveries are post-origination outcomes and were removed from the PD feature set; they are used only to construct LGD/EAD targets. Corrected metrics (KS 31) replace the pre-fix leakage figures (KS 80).

CHAMPION CHOICE Challenger lift is only +0.013 Gini / +1.0 KS - insufficient to forgo the scorecard's full explainability. The logistic scorecard is retained as the production champion.

LGD / EAD EAD $R^2=0.79$ (strong). LGD $R^2=0.02$ - expected: recoveries on unsecured loans are near-zero, so LGD clusters at ~0.92. Use portfolio-mean LGD as a sanity floor.

RESIDUAL RISK grade / sub_grade / int_rate are collinear (grade insignificant, $p=0.27$). Retained for transparency; recommend keeping a single pricing variable at next redevelopment.

5. Ongoing monitoring (PSI, dev vs out-of-time)

Output-score PSI = 0.0059 -> STABLE (thresholds 0.10 / 0.25).
Feature PSI: 0 critical, 2 minor shift (worst: bin_bc_util = 0.153), rest stable.
Cadence: recompute monthly; escalate to recalibration if any PSI > 0.25 or fresh-vintage KS < 30.

VALIDATION OPINION: APPROVED FOR DEPLOYMENT (with conditions)

Model passes the $KS \geq 30$ discrimination gate and is stable out-of-time. Conditions: monthly PSI monitoring; resolve pricing-variable collinearity and adopt a downturn-LGD floor at next scheduled redevelopment.